

LUDO GAME — SIMPLE DETAILED DOCUMENTATION

Rule Version: Standard Classic
Triple Six Rule: Enabled
Block Behaviour: Protective (No Barrier)
Block Crossing: Allowed
Opponent Landing on Block: Allowed
Final Home Bonus Turn: Enabled.

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1. Goal of the Game

Ludo is a board game where:

- Each player has 4 pieces
- Players take turns rolling a dice
- Pieces move around the board
- The first player to bring all 4 pieces to the center (home) wins

2. Number of Players

The game supports:

- Minimum: 2 players
- Maximum: 4 players
- Each player gets one color:
 - Red
 - Blue
 - Green
 - Yellow

3. Game Setup

When a new game starts:

- All players join the room
- Each player gets a color
- Each player has 4 pieces
- All pieces start inside the player's base
- One player is randomly chosen to start first

Game status becomes: **started**

4. Basic Turn Flow

Each player's turn follows the same pattern:

- Player rolls the dice
- System checks which pieces can move
- Player chooses a piece
- Piece moves
- System checks special rules
- Turn ends or continues

❖ This repeats until the game finishes.

5. Dice Rules

5.1 Dice Range

The dice gives a number between **1 and 6**.

5.2 When Player Rolls a 6

If a player rolls 6:

- They may bring a new piece out of the base
- OR move an existing piece
- AND they usually get another turn

Important: Extra turn only happens if a move was actually made.

5.3 When Player Cannot Move

Sometimes after rolling the dice:

- No piece can move
- All pieces are blocked

In this case:

- The turn automatically ends.

6. Moving Pieces

6.1 Bringing Piece Out of Base

A piece can leave the base only when:

- Player rolls a 6

When this happens:

- The piece is placed on the starting square of that color
- This counts as a move

6.2 Moving on the Board

If a piece is already on the board:

- It moves forward by the dice number
- Movement is always in a fixed path
- Pieces cannot move backward

6.3 Entering the Home Path

After completing almost one full round:

- The piece enters its color's home path
- Only that player can use that path

6.4 Reaching Final Home

To reach the final center:

- The player must roll the exact number
- If the dice number is too big, the piece cannot move

7. Cutting Opponent Pieces

7.1 What is Cutting?

If your piece lands on a square where an opponent's piece is standing:

- The opponent's piece is sent back to its base.

This is called cutting.

7.2 Bonus for Cutting

When a player cuts an opponent:

- The player gets one extra turn

7.3 Safe Squares (Stars)

There are some special safe squares.

If a piece stands on a safe square:

- It cannot be cut
- Multiple players can stand there safely

8. Same Color Pieces Together(Block Rule)

8.1 Creating a Block

If two pieces of the same color stand on the same square:

- They form a block

8.2 Block Protection

- A block cannot be cut
- If an opponent lands on that square, the block remains safe
- Landing on that square does NOT send the block pieces to base

8.3 Movement Around Block

- Opponents are allowed to land on that square
- Opponents are allowed to cross that square
- The block does not stop movement

9. Extra Turn Rules

A player gets another turn if:

- They rolled a 6 and made a move
- They cut an opponent
- Their piece reached the final home (optional rule)

10. Turn Ending

The turn passes to the next player when:

- No extra turn is earned
- Player cannot move
- Player finishes their allowed actions

The next player is chosen in clockwise order.

11. Winning the Game

11.1 When Player Finishes

A player finishes when:

- All 4 of their pieces reach the final home

Their finishing position is recorded.

11.2 When Game Ends

Game ends when:

- Only one player remains unfinished
OR
- All players finish

12. Player Timeout (Important)

If a player does not act within the time limit:

Possible actions:

- Skip their turn (recommended)
- Let computer play for them
- Pause the game

Recommended: **auto-skip after fixed time**

13. Player Disconnect Handling

If a player loses internet:

When they return, they should see:

- Current board
- Whose turn it is
- Dice value (if already rolled)
- All piece positions

Game must continue smoothly.

14. Important Edge Cases

These must be handled carefully.

Edge Case 1: Rolled 6 but No Move Possible

Rule:

1. Player does NOT get extra turn
2. Turn ends

Edge Case 2: Exact Number for Home

If the piece needs 3 but dice shows 5:

1. Move is not allowed

Edge Case 3: Multiple Pieces Possible

If many pieces can move:

1. Player must choose one
2. System must wait for choice

Edge Case 4: All Pieces in Base and No 6

1. Player cannot move
2. Turn ends automatically

Edge Case 5: Duplicate Actions

System must prevent:

- Rolling dice twice
- Moving twice

- Playing out of turn

Edge Case 6: Game Stuck Situation

Very rare but possible.

System should ensure:

1. Game always moves to next player
2. No infinite waiting

15. Recommended Game Records

For fairness and debugging, store:

- Dice history
- Move history
- Cut events
- Winner order
- Turn times

16. Board Structure Details

16.1 Main Board Path

- The main path contains total 52 squares(Each player travels 51 squares before entering home)
- All players move on the same main path
- Each color has a fixed starting square
- Movement direction is clockwise

16.2 Starting Squares

- Red has its own starting square
- Blue has its own starting square
- Green has its own starting square
- Yellow has its own starting square
- A piece enters the board only on its color's starting square

16.3 Home Path

- Each player has a separate home path
- Home path contains 6 squares
- Only that color can enter its home path
- Other players cannot enter it

16.4 Final Home Position

- The last square of the home path is the final home
- Exact dice number is required to reach it

17. Triple Six Rule

17.1 Consecutive Six Rolls

If a player rolls 6 three times in a row:

- The turn is cancelled
- No piece moves for the third roll
- No extra turn is given
- Turn passes to the next player

(If this rule is not supported, clearly mention that it is disabled.)

18. Cutting Restrictions

18.1 Cutting Area

- Cutting is allowed only on the main path
- Cutting is NOT allowed inside home path
- Cutting is NOT allowed on safe squares

18.2 Multiple Pieces on Same Square

If more than one opponent piece is standing (if allowed by your rule):

- All pieces are sent back to base

19. Ranking Rules

19.1 First Winner

- First player to finish all 4 pieces gets Rank 1

19.2 Second and Third

- Ranking is decided by order of completion

19.3 Final Player

- The last unfinished player automatically gets last rank

20. Turn Timer Rules

20.1 Dice Roll Time

- Player must roll dice within fixed time (example: 15 seconds)

20.2 Piece Selection Time

- After rolling, player must choose piece within fixed time

20.3 Timeout Action

If player does not act in time:

- Turn is automatically skipped

Recommended:

Use same timer duration for all players

21. Game Restart Rules

21.1 After Game Ends

- Final ranking is shown
- Match summary is displayed

21.2 Rematch Option

Players may:

- Start a new game with same players
- OR
- Exit the room

21.3 New Game Reset

When new game starts:

- All pieces return to base
- New first player is chosen randomly

22. Spectator Rules (Optional)

22.1 Watching the Game

- Spectators can watch the board
- Spectators cannot roll dice
- Spectators cannot move pieces

22.2 Joining Mid-Game

Recommended rule:

- New players cannot join after game starts

23. Anti-Cheat Protection Rules

System must prevent:

- Rolling dice when not player's turn
- Moving piece without rolling
- Moving more than dice value
- Moving opponent's piece
- Performing multiple actions in same turn

All actions must be validated by system.

24. Deadlock Prevention Rule

Very rare situation:

If the game continues for very long without progress:

Recommended handling:

- Continue normal gameplay
- OR
- Apply maximum turn limit (example: 300 turns)

After limit:

- Player with most pieces in home wins
- OR
- Game is declared draw

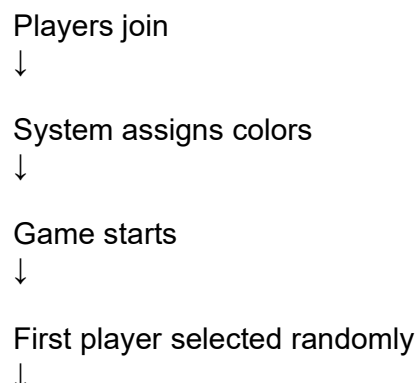
25. Game Progress Records

For transparency and debugging, store:

- Total number of turns
- Total number of cuts
- Total number of sixes rolled
- Time taken by each player
- Final ranking order

26. Full Game Flow Summary

Simple overview:



Start turn timer



Player rolls dice



Check for triple six rule



System checks possible moves



If no move possible



Turn ends → Next player

If move possible



Player selects piece



System validates move



Piece moves



Check block rule



Check cut (main path only)



If opponent cut → Send piece to base



Check if piece entered home path



Check if piece reached final home (exact number required)

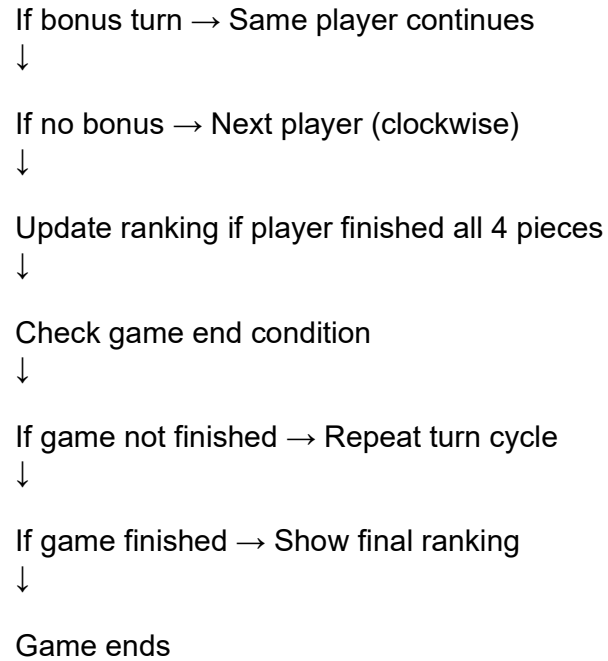


Update player finished pieces count



Check bonus turn condition
(roll 6 / cut / home reach)





THANK YOU !!